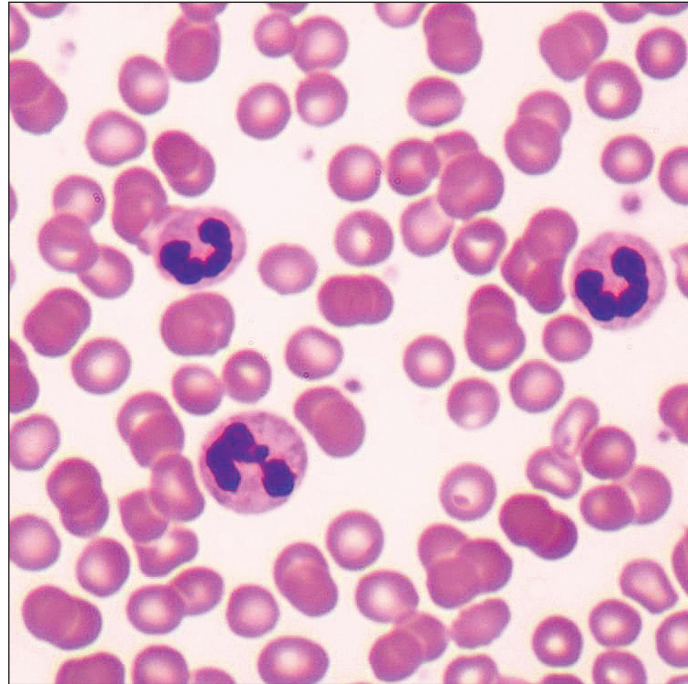


3. **Image 3.1** shows a sample of blood viewed down a light microscope.

Image 3.1



(a) **Draw labelled arrows** on **Image 3.1** to show:

[1]

- **one** red blood cell
- **one** white blood cell.

(b) (i) The blood of a human male contains 43 cm^3 of plasma per kg of body mass.
The volume of blood in a human male of mass 62 kg is 4670 cm^3 .
Calculate the percentage of plasma in the blood of this male.

Write your answer to two significant figures.

[3]

Percentage =

(ii) State **one** function of plasma.

[1]

.....



- (c) Health professionals can use a full blood count (FBC) to monitor health. **Table 3.2** shows some values from an FBC for three patients, **A**, **B** and **C**. The normal ranges are also shown.

Table 3.2

Component of blood	Concentration in blood (a.u.)			
	Normal range	Patient A	Patient B	Patient C
Haemoglobin	13.2–14.8	10.9	13.5	13.2
Red blood cells	4.5–5.2	4.4	4.6	5.1
White blood cells	2.0–7.0	5.9	0.4	1.3
Platelets	199.2–292.1	234.4	313.3	42.4

Use **Table 3.2** and your knowledge to explain:

- (i) why patient **A** is most likely to suffer low blood oxygen levels; [2]

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- (ii) why patient **B** is most at risk of infectious disease. [2]

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